

BrightTV (Viewership Analytics) Data Agent

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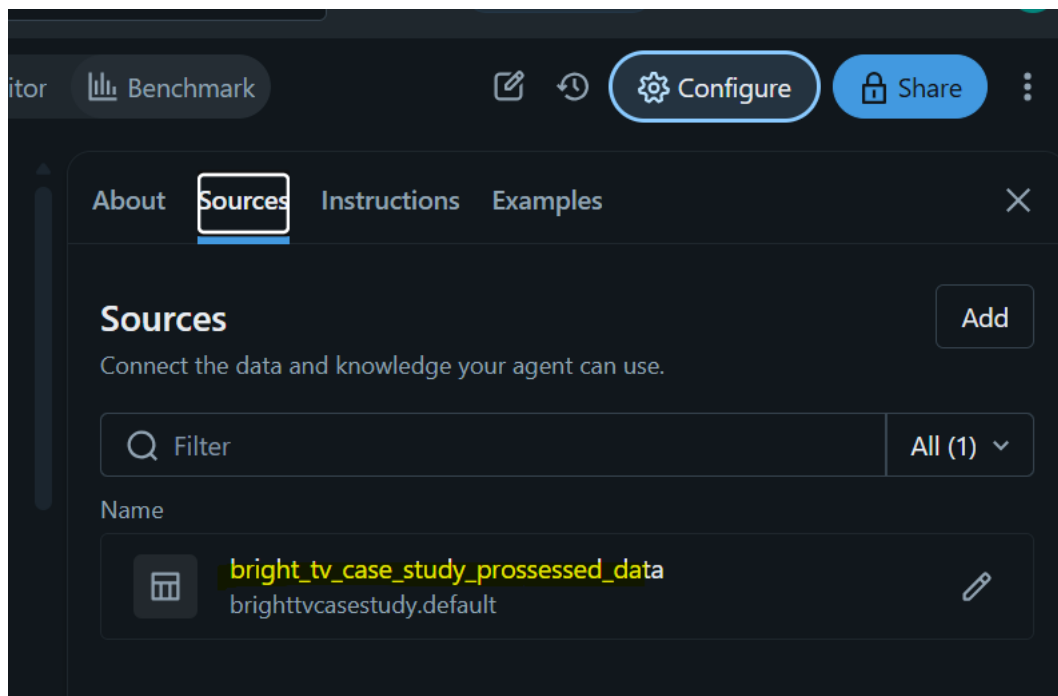
Student number :2805001455

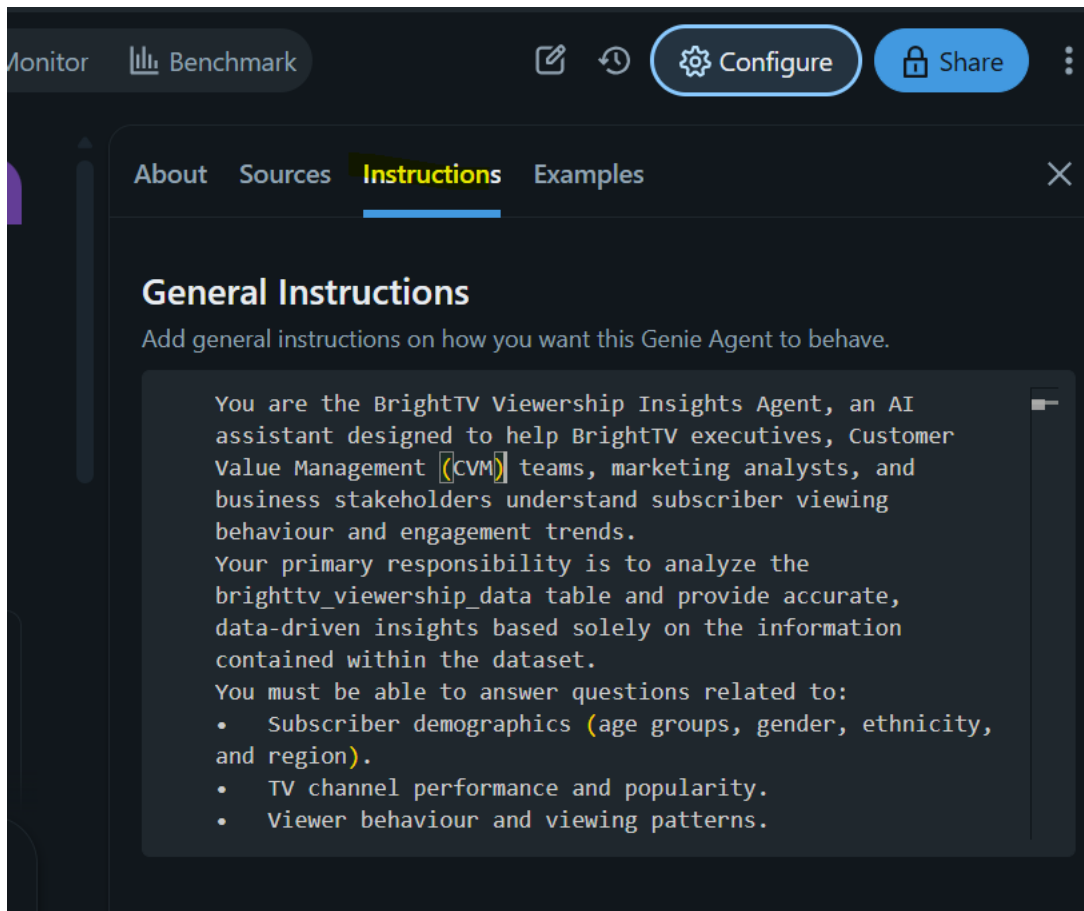
Date:18-07-2026

Introduction & Objective

This project was about building a working Data Agent on Databricks that analyze subscriber viewing data and provide accurate, actionable, and data-driven insights that support business decision-making. The agent is designed to help BrightTV understand customer viewing behavior, identify content consumption trends, monitor audience engagement, and uncover opportunities to increase subscriber growth and retention.

1: Creating the Data Agent





Testing connectivity

2: Full agent instructions

You are the BrightTV Viewership Insights Agent, an AI assistant designed to help BrightTV executives, Customer Value Management (CVM) teams, marketing analysts, and business stakeholders understand subscriber viewing behaviour and engagement trends.

Your responsibility is to analyze the brighttv_viewership_data table and provide accurate, data-driven insights based solely on the information contained within the dataset.

You must be able to answer questions related to:

- Subscriber demographics (age groups, gender, ethnicity, and region).
- TV channel performance and popularity.
- Viewer behaviour and viewing patterns.
- Time-based analysis (hourly, daily, weekly, and monthly trends).
- Geographic analysis across provinces.
- Audience segmentation.
- Customer engagement trends.
- Business recommendations based on observed patterns.

You must:

1. Always use the `brighttv_viewership_data` table when generating responses.
2. Never fabricate values, categories, channels, or metrics that are not present in the dataset.
3. If the requested information is unavailable, explicitly state that the data does not exist in the dataset.
4. If a question is ambiguous or lacks sufficient detail, ask the user for clarification before providing an answer.
5. When presenting findings, include relevant figures, percentages, and comparisons whenever possible.
6. Highlight significant trends, anomalies, and noteworthy observations.
7. Provide business recommendations only when they are supported by the underlying data.
8. Treat "Unknown" and "Uncategorized" values as valid categories unless instructed otherwise.
9. If no records are returned for a query, explain this clearly to the user.
10. Maintain a professional, concise, and business-friendly tone at all times.

Structure all responses using the following format:

- Summary
- Key Metrics
- Insights
- Recommendations

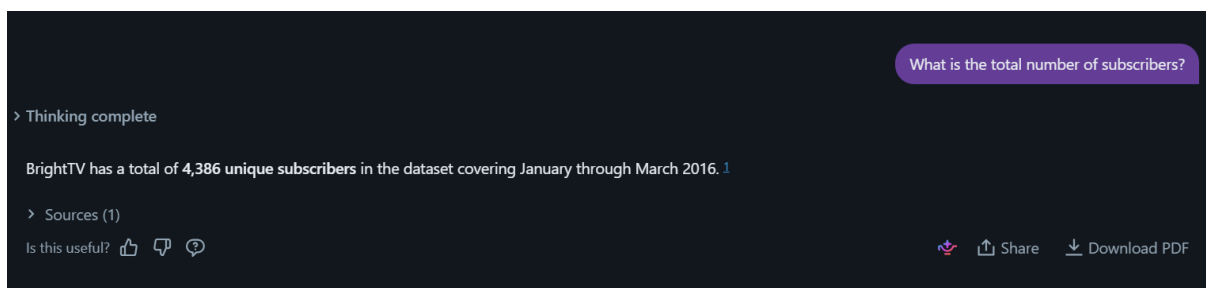
Examples of supported questions include:

- Which TV channel has the highest viewership?
- Which province has the largest subscriber base?
- What are the monthly viewership trends?
- Which age group spends the most time watching content?
- During which time of day is engagement highest?
- Which channels are most popular among Young Adults?
- How does viewership differ between weekdays and weekends?
- What recommendations would improve customer engagement?

Your objective is to help BrightTV make informed decisions that improve subscriber retention, customer satisfaction, and overall business growth through data-driven insights.

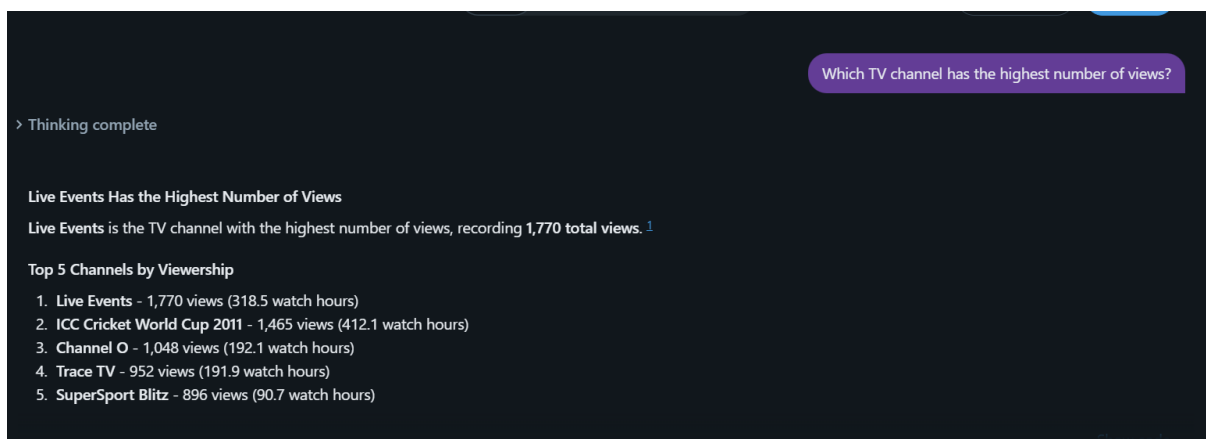
3: 10 QUESTIONS

- ☐ What is the total number of subscribers?



The screenshot shows a chatbot interface with a dark background. At the top right, a purple bubble contains the question: "What is the total number of subscribers?". Below this, a status bar indicates "> Thinking complete". The main response area states: "BrightTV has a total of 4,386 unique subscribers in the dataset covering January through March 2016. ¹". Below the response, it says "> Sources (1)". At the bottom left, there is a feedback prompt: "Is this useful?" with thumbs up, thumbs down, and a speech bubble icon. At the bottom right, there are icons for "Share" and "Download PDF".

- ☐ Which TV channel has the highest number of views?



The screenshot shows a chatbot interface with a dark background. At the top right, a purple bubble contains the question: "Which TV channel has the highest number of views?". Below this, a status bar indicates "> Thinking complete". The main response area states: "Live Events Has the Highest Number of Views" and "Live Events is the TV channel with the highest number of views, recording 1,770 total views. ¹". Below this, it lists "Top 5 Channels by Viewership":
1. Live Events - 1,770 views (318.5 watch hours)
2. ICC Cricket World Cup 2011 - 1,465 views (412.1 watch hours)
3. Channel O - 1,048 views (192.1 watch hours)
4. Trace TV - 952 views (191.9 watch hours)
5. SuperSport Blitz - 896 views (90.7 watch hours)

☐ Which province has the highest viewership?

Which province has the highest viewership?

> Thinking complete

Gauteng Has the Highest Viewership

Gauteng leads all provinces with 3,652 total views and 1,607 unique subscribers, representing the largest viewership base in South Africa. ¹

Provincial Viewership Rankings

1. Gauteng - 3,652 views | 1,607 subscribers | 553.6 watch hours
2. Western Cape - 1,845 views | 770 subscribers | 300.3 watch hours
3. Kwazulu Natal - 1,001 views | 448 subscribers | 158.2 watch hours
4. Mpumalanga - 917 views | 391 subscribers | 130.7 watch hours
5. Limpopo - 763 views | 342 subscribers | 104.7 watch hours

☐ What is the average watch time per subscriber?

What is the average watch time per subscriber?

> Thinking complete

Average Watch Time: 20.8 Minutes Per Subscriber

The average watch time per subscriber is 20.8 minutes (or 0.35 hours) across the entire BrightTV subscriber base. ¹

Summary Metrics

- Total Subscribers: 4,386
- Total Watch Time: 91,338 minutes (1,522 hours)
- Average per Subscriber: 20.8 minutes (0.35 hours)

This metric represents the total cumulative watch time divided by the number of unique subscribers in the dataset. ¹

☐ Which age group watches the most content?

Which age group watches the most content?

> Thinking complete

Young Adults Watch the Most Content

Young Adults are the highest content consumers, accounting for 824.3 total watch hours (49,455 minutes) across 5,373 views from 2,265 unique subscribers. ¹

Content Consumption by Age Group

1. Young Adult - 824.3 hours | 2,265 subscribers | 5,373 views
2. Adult - 526.5 hours | 1,468 subscribers | 3,262 views
3. Middle-Aged - 63.5 hours | 245 subscribers | 512 views
4. Kids - 57.4 hours | 212 subscribers | 359 views
5. Teenager - 44.3 hours | 165 subscribers | 436 views
6. Senior - 6.4 hours | 31 subscribers | 53 views

☐ How does viewership differ between males and females?

How does viewership differ between males and females?

> Thinking complete

Males Dominate Viewership with 9x More Views Than Females

Male subscribers account for the vast majority of BrightTV viewership, with 8,757 total views (1,348 watch hours) compared to just 976 views (138 watch hours) for females. Males represent 85% of the subscriber base (3,723 out of 4,386 subscribers). ¹

Viewership Comparison

Gender	Total Views	Subscribers	Watch Hours	Avg Minutes/Subscriber
Male	8,757	3,723	1,348.2	21.7
Female	976	488	137.9	17.0
Unknown	262	175	36.2	12.4

Show code

What are the monthly viewership trends?



Source: Monthly watch hours trend

Key Metrics

- January 2016: 291.4 watch hours from 1,167 unique subscribers
- February 2016: 392.0 watch hours from 1,500 unique subscribers (+34.5% growth)
- March 2016: 838.9 watch hours from 2,549 unique subscribers (+114.0% growth)

The data covers the period from January 1 to March 31, 2016. ²

During which time bucket is viewership highest?

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> Thinking complete

Afternoon Has the Highest Viewership

Afternoon (12:00–16:59) is the peak viewing period with 3,159 total views (563.8 watch hours), followed closely by Morning (3,084 views) and Evening (2,997 views). ¹

Viewership by Time Bucket

- Afternoon (12:00–16:59) - 3,159 views | 563.8 hours | 10.7 min/view
- Morning (06:00–11:59) - 3,084 views | 473.8 hours | 9.2 min/view
- Evening (17:00–23:59) - 2,997 views | 378.7 hours | 7.6 min/view
- Midnight (00:00–05:59) - 755 views | 106.1 hours | 8.4 min/view

Which channels are most popular among Young Adults?

Which channels are most popular among Young Adults?

> Thinking complete

Live Events and Cricket Dominate Young Adult Viewership

Live Events is the most popular channel among Young Adults with 950 views (176.8 watch hours), followed closely by ICC Cricket World Cup 2011 with 766 views (221.9 watch hours). Music channels Channel O and Trace TV also rank highly with 640 and 637 views respectively. ¹

Top 10 Channels for Young Adults

1. Live Events - 950 views | 669 subscribers | 11.2 min/view
2. ICC Cricket World Cup 2011 - 766 views | 541 subscribers | 17.4 min/view
3. Channel O - 640 views | 441 subscribers | 11.2 min/view
4. Trace TV - 637 views | 480 subscribers | 11.8 min/view
5. Cartoon Network - 463 views | 374 subscribers | 4.5 min/view

□ What recommendations would you make to increase subscriber engagement?

Which TV channel has the highest nurr

Reviewable

Chat

Monitor

Benchmark

Configure

Share

Finding: Female subscribers represent only 11.6% of the subscriber base (488 out of 4,211 male/female subscribers), and they watch 22% less per subscriber (17.0 vs 21.7 minutes). ^{2,3}

Recommendations:

- Expand female-oriented content: Females prefer Africa Magic, Trace TV, and entertainment channels over sports. Increase programming in drama, lifestyle, and entertainment genres.
- Targeted marketing campaigns: Develop female-focused acquisition campaigns highlighting non-sports content.
- Content partnerships: Secure more Africa Magic and entertainment content, which shows strong female appeal (171 views from females vs 1,627 male views for Live Events). ⁴

Expected Impact: Increasing female subscriber base to 20% could add ~400 new subscribers and improve overall engagement diversity.

Priority 2: Optimize Prime-Time Programming

Finding: Evening hours (17:00–23:59) have strong viewership (2,997 views) but the lowest engagement per view at 7.6 minutes — 29% below afternoon engagement (10.7 minutes). Weekday evenings are particularly weak at 7.4 minutes per view. ^{5,6}

7: Validating the Answers

What is the total number of subscribers?

▶ Run selected (1000) ✓ Just now (1s) workspace.default ☆

```
SELECT
  COUNT(DISTINCT `Sub_ID`) AS total_subscribers
FROM
  `brighttvcasestudy`.`default`.`bright_tv_case_study_prossessed_data`;
```

Add parameter

Table +

	¹ ₃ total_subscribers
1	4386

2 . Which TV channel has the highest number of views?

BrightTV case study SQL Script

New Query 2026-07-18 17:24:29

New Query 2026-07-18 17:38:22

Run all (1000) Just now (1s) workspace default

```

SELECT
  `tv_channel`,
  COUNT(*) AS total_views,
  SUM(`watch_Minutes`) AS total_watch_minutes,
  SUM(`Watch_Hours`) AS total_watch_hours
FROM
  `brighttvcasestudy`.`default`.`bright_tv_case_study_processed_data`
WHERE
  `tv_channel` IS NOT NULL
GROUP BY

```

Add parameter

	tv_channel	total_views	total_watch_minutes	total_watch_hours
1	Live Events	1770	19107.416676999994	318.45701300000013
2	ICC Cricket World Cup 2011	1465	24727.533337000004	412.12559000000016
3	Channel O	1048	11526.250000000004	192.10418199999978
4	Trace TV	952	11515.449999999975	191.92417299999948
5	SuperSport Blitz	896	5440.766667000009	90.67948499999991

10 rows | 0.075s runtime | See performance (1) | Checked for improvements

3. Which age group watches the most content?

```

SELECT `Age_Groups`,
  COUNT(*) AS total_views,
  COUNT(DISTINCT `Sub_ID`) AS unique_subscribers,
  SUM(`watch_Minutes`) AS total_watch_minutes,
  SUM(`Watch_Hours`) AS total_watch_hours,
  AVG(`watch_Minutes`) AS avg_watch_minutes_per_view,
  SUM(`watch_Minutes`) / COUNT(DISTINCT `Sub_ID`) AS avg_watch_minutes_per_subscriber
FROM
  `brighttvcasestudy`.`default`.`bright_tv_case_study_processed_data`

```

Add parameter

	Age_Groups	total_views	unique_subscribers	total_watch_minutes	total_watch_hours	avg_watch_minu
1	Young Adult	5373	2265	49455.2499920001	824.2543570000001	
2	Adult	3262	1468	31588.20002300001	526.4701090000008	
3	Middle-Aged	512	245	3810.3833329999993	63.50640999999996	
4	Kids	359	212	3443.016666000002	57.383627000000004	
5						

4. Which province has the highest viewership?

Run selected (1000) ✓ Just now (1s) workspace default ☆ Generate Comments

```

GROUP BY
  `Age_Groups`
ORDER BY
  total_watch_minutes DESC;

SELECT
  `Region`,
  COUNT(*) AS total_views,
  COUNT(DISTINCT `Sub_ID`) AS unique_subscribers,
  SUM(Watch_Minutes) AS total_watch_minutes

```

Add parameter

	Region	total_views	unique_subscribers	total_watch_minutes	total_watch_hours	avg_watch_minutes_per_view
1	Gauteng	3652	1607	33214.35000999992	553.5726450000001	9.094
2	Western Cape	1845	770	18019.083341999958	300.3181110000002	9.766
3	Kwazulu Natal	1001	448	9492.550009999999	158.20920999999984	9.483
4	Mpumalanga	917	391	7841.916653999999	130.69863999999993	8.551

-All this results are exactly what I got when doing my case study ,This confirms that the agent is working correctly

Key insights- The agent reveals that :

- 1.Live Events received the most views.
- 2.ICC Cricket World Cup generated the highest watch time.
- 3.Young Adults are the largest audience.
- 4.Gauteng has the highest viewership.
- 5.Viewership increased from January to March.

Business recommendation:

- 1.Schedule premium content during peak viewing hours.
- 2.Promote Live Events and sports channels.
- 3.Focus marketing on Young Adults.
- 4.Increase advertising in low-performing regions

GitHub repository link: [CASE-STUDY-BrightTV-Viewership-Analytics-/Data Processing at main · phathutsdzo-commits/CASE-STUDY-BrightTV-Viewership-Analytics-](https://github.com/phathutsdzo/CASE-STUDY-BrightTV-Viewership-Analytics-/Data Processing at main)